Assignment I - Kelly Noble 9/24/26

Grade: REVIEW_REQUIRED

Review Notes:

x Qa: Review manually.
x Qb: Review manually.
x Qc: Review manually.
x Qd: Review manually.
x Qe: Review manually.

Question A: Withdraws at least $60.

10	0.10
20	0.15
40	0.30
60	0.15
	0.70

sum

70% probability customer withdraws at least $60

Question B: EV w/drawn is $50 ± $26.27

x		f(x)		x · f(x)
10	×	0.10	=	1.0
20	×	0.15	=	3.0
40	×	0.30	=	12.0
60	×	0.15	=	9.0
80	×	0.25	=	20.0
100	×	0.05	=	5.0

sum

$\boxed{\$50 = E(x)}$ Expected w/drawl amt.

Standard Deviation $\mu = \$50$ $Var(x) = \sum (x-\mu)^2 f(x)$

x	f(x)	x − μ	(x − μ)²	(x − μ)² · f(x)
10	0.10	−40	1600	160
20	0.15	−30	900	135
40	0.30	−10	100	30
60	0.15	10	100	15
80	0.25	30	900	225
100	0.05	50	2500	125
				690

sum

$Var(x) = 690$ $\boxed{\sigma = \sqrt{690} = \$26.27}$